

FILTER DESIGN TEST CASE

TC-SAW (Temperature Compensated SAW) - WiFi 6E (6 GHz) for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0071
Revision:	Rev A.8
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Created Date:	2024-11-08
Last Modified:	2024-11-24
Status:	NEEDS REVIEW
Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a TC-SAW (Temperature Compensated SAW) filter targeting WiFi 6E (6 GHz) for Satellite L-Band Filter. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 5342.7 MHz and bandwidth of 33.5 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: TC-SAW (Temperature Compensated SAW)
- Frequency Band: WiFi 6E (6 GHz)
- Application: Satellite L-Band Filter
- Substrate: 36° YX LiTaO3
- Package: Flip-Chip BGA
- Design Tool: CST Studio Suite 2024
- Design Center: Singapore RF Lab

This specification applies to all variants of the TC-SAW (Temperature Compensated SAW) design targeting WiFi 6E (6 GHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5342.7 MHz
Bandwidth (3 dB):	33.5 MHz
Insertion Loss Target:	≤ 1.8 dB
Return Loss Target:	≥ 9.0 dB
Out-of-Band Rejection:	≥ 31.0 dB
Isolation (Duplexer):	≥ 33.0 dB
Q Factor:	11311
Temperature Coefficient:	-15.1 ppm/°C
Die Size:	2.1 x 2.2 mm
Wafer Size:	6-inch
Number of Resonators:	7
Electrode Material:	Mo

Passivation: SiO2

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the TC-SAW (Temperature Compensated SAW) template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 5342.7 MHz, BW 33.5 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.8 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 16028 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.8 dB
- Return loss in passband shall be >= 9.0 dB
- Out-of-band rejection at +/- 50 MHz offset >= 31.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 10.1 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 91%

6. TEST RESULTS

Measurement Date: 2024-11-24
Devices Tested: 35
Insertion Loss (meas): 1.54 dB
Return Loss (meas): 10.9 dB
Rejection (meas): 34.1 dB
Isolation (meas): 37.6 dB
Q Factor (meas): 11311
Group Delay Variation: 7.0 ns
Power Handling: +36 dBm
Die Yield: 96.7%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The TC-SAW (Temperature Compensated SAW) design demonstrated below-target performance for WiFi 6E (6 GHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Group delay flatness improved compared to previous revision.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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